Yuzhu Tao

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Research Interests

Computer vision, image stitching, image editing models, and multimodal large language models.

Education

Beijing Jiaotong University, M.S. in Artificial Intelligence — Beijing, China — Sep 2024 - Jun 2027

- Research direction: image stitching in complex scenes.

Hebei Normal University, B.S. in Computer Science and Technology — Hebei, China — Sep 2020 - Jun 2024

- Weighted Average: 90.45; Rank: 1/147.
- Key Courses: Advanced Mathematics 98, Probability and Statistics 94, Digital Image Processing 95.
- College English Test Band 6: 453.

Publications

Fast-Rotating Panoramic HDR Reconstruction from Single-Second Spikes, *Yuzhu Tao*, Yakun Chang, Shikui Wei, Lang Nie, Qingwen Li, Bo Li, Boxin Shi, Yao Zhao — ACM Multimedia (CCF-A) — 2026

- Problem: capturing panoramic HDR images during fast rotation is challenging due to motion blur and overexposure.
- Solution: proposed a side-by-side hybrid camera system requiring no spatial/temporal alignment; designed a spike-based panoramic reconstruction method and a colorization module based on a fine-tuned multimodal large language model; built a comprehensive dataset with synthetic and real-world spikes for HDR panoramic reconstruction and colorization.
- Result: reconstructed full-color panoramic HDR images from one-second spike acquisition under fast-rotation conditions.

Seam-Guided Unsupervised Image Stitching with Parallax-Aware Mask Generation, *Yuzhu Tao*, Lang Nie, Yakun Chang, Bo Li, Qingwen Li, Hong Li, Shikui Wei — IEEE TCSVT (CCF-B) — 2026

- Problem: image stitching under large parallax is challenging due to the conflict between content alignment and shape preservation; spatially varying transformations often cause unexpected distortions.
- Solution: designed an edge-enhanced mask generation module to distinguish large-parallax regions and suppress excessive distortions; proposed a seam-guided alignment strategy that iteratively refines local registration for improved stitching consistency; built a new parallax dataset to improve generalization.
- Result: effectively removes misaligned regions while maintaining shape consistency, especially in challenging parallax scenarios.

Internship Experience

Baidu Online Network Technology Co., Ltd., Multimodal Algorithm Research Engineer — China — Mar 2026 - Aug 2026

- Background: conducted algorithm research on image understanding and image editing generation.

Project 1: Large Multimodal Models for Image Understanding, Intelligent Portrait Retouching Parameter Prediction

- Problem: intelligent portrait retouching requires a vision-language model to perceive facial flaws and predict structured retouching parameters, replacing inefficient manual tuning.
- Solution: fine-tuned Qwen3-VL-4B with LoRA for multi-label retouching parameter prediction (input: face image → output: structured JSON of retouching functions and levels). **Stage 1 (SFT):** trained on Gemini-labeled instruction data for visual perception and parameter prediction; **Stage 2 (paired-data SFT):** built a data flywheel — used Stage-1 model to infer parameters on 160K images, applied AI skills retouching tools to generate before-after pairs as augmented training data.
- Result: on a 900-image test set, improved prediction F1 from 81% to 86%.

Project 2: Creative Image Generation, AI ID-Photo Outfit Replacement

- Problem: AI ID-photo outfit replacement fails when the subject's head and neck are misaligned or the neck region is missing, producing unnatural results.

- Preprocessing: extracted human keypoints to align head and neck before generation.
- Solution: **Stage 1 (Data Construction):** used Nano Banana 2 to construct head-neck images, applied SAM3 for neck segmentation, and simulated structural missing regions via random masking to build training data. **Stage 2 (Neck Inpainting):** fine-tuned Qwen-Image-Edit-2509 as an image editing model to regenerate the missing neck structure, completing the full portrait from incomplete inputs. **Stage 3 (Post-processing):** applied texture phase-shift overlay to mitigate skin over-smoothing and used keypoints to place clothing.
- Result: achieved natural AI ID-photo outfit replacement with correct neck structure and realistic skin texture.

Projects

Spherical-Projection Panoramic Image Stitching System for Data Centers, Core Algorithm Engineer — Beijing, China — Sep 2024 - Sep 2025

- Problem: a single camera has limited FOV in data-center scenes, requiring panoramic reconstruction from multiple views.
- Solution: proposed a fast spherical-projection panoramic stitching algorithm — corrected lens distortion, estimated focal length via cylindrical projection, built a spherical projection model to fuse 30 images into an equirectangular projection (ERP) panorama supporting 3D viewing.
- Result: reconstructs a full panorama in one second; deployed in China Tower's production business; authored **one patent** (owned by China Tower).

Patents

High-Speed High-Dynamic-Range Color Panoramic Image Reconstruction Method and System (Chinese Invention Patent) — 2026

- Patent application number: 2026100594905.

Honors and Awards

First-Class Scholarship for Academic Excellence of Beijing Jiaotong University, 2024, 2025.

Outstanding Student Leader, Beijing Jiaotong University, 2025.

Outstanding Graduate Communist Party Member, Beijing Jiaotong University, 2025.

Outstanding Student, Beijing Jiaotong University, 2025.

Outstanding Undergraduate Graduate of Hebei Province, 2024.

Second Prize, Lanqiao Cup C/C++ Programming Competition, 2023.

Honorable Mention, Hebei Provincial Collegiate Programming Contest, 2023.

Third Prize, National College Students Mathematics Competition, 2023.

First-Class Scholarship for Academic Excellence of Hebei Normal University, 2021, 2022, 2023.

Outstanding Student, Hebei Normal University, 2021, 2022.

Outstanding Student Leader, Hebei Normal University, 2021.

Skills

Programming Languages: Python, C

Deep Learning: PyTorch, Transformers, Diffusers

Computer Vision: image stitching, image editing

Multimodal Models: image editing models, multimodal large language models

Research Statement

My research lies at the intersection of geometric computer vision and generative image modeling. Through my stitching work (TCSVT) and panoramic reconstruction (ACM MM), I identified large parallax as a fundamental limitation of alignment-based methods, and I am interested in exploring image editing models for local parallax region regeneration. During my internship, I observed shared texture fidelity challenges across tasks: VAE decoders losing high-frequency detail (ID-photo generation), tiling artifacts in repetitive textures like grass, and the inability of super-resolution models to recover semantic information. These observations motivate my PhD research interest: understanding and solving texture-faithful generative editing under geometric constraints.